

CSE340: Computer Architecture

Assignment -1

Deadline: 28th February, 3 AM

Assignment must be handwritten. Scan and upload PDF in provided Google form

Question - 1:

Why is it important to keep redundancy while designing a system? **Explain** a scenario where this redundancy will be useful.

Question - 2:

You are given a table with reference times and measured execution times for 4 benchmark programs.

Program Name	Execution Time (seconds)	Reference Time (seconds)
Perl bench	200	250
x264	120	180
gcc	400	300
leela	60	90

- To calculate the **benchmark** of a system, why do we take the **geometric mean** instead of only taking the **average** of the individual spec ratios?
- What does a SPEC ratio of greater than 1 signify?
- Calculate the SPEC ratio of a new CPU Z and explain what it means.

Question - 3:

You are a hardware engineer at a gaming laptop company. You must choose between two CPUs for a high-performance model.

CPU Model	CPI	Clock Cycle Time
Alpha	3.2	2 ns
Beta	2.8	2.5 ns

A benchmark test program runs **1.5 billion instructions**.

- Calculate the CPU time for both CPUs?
- Which CPU performs better and by how much?
- Considering the gaming workload needs **low latency**, which CPU would you recommend?

Question - 4:

- Explain** Amdahl's Law in your own words.
- Can you relate/connect Amdahl's law with any of the design principles, mentioned below? Explain the reason with an example.

	Design Principle
1	Performance via Prediction
2	Performance via Pipelining
3	Make the common case faster
4	Use Abstraction to Simplify Design

- You want to optimize a video rendering software. It currently takes 10 seconds to complete rendering, of which 40% of the time is spent on color correction. If you improve the task of color correction by 4 times, what would be the new rendering time of the software? How much of an improvement is it compared to the initial time?

Question - 5:

Consider you are comparing two different INTEL CPUs X and Y. Both of these CPUs are running a program that supports 3 types of instructions A, B, and C.

CPU	Clock Rate (GHz)	Class A CPI	Class B CPI	Class C CPI
X	3.2	3	4	2
Y	3.6	2	3	2

The program has a total instruction count of 5×10^5 divided in the following ratios:

Instruction Class	Percentage of Instructions
A	40%
B	40%
C	20%

Now answer the following:

1. Calculate the **average CPI** for both CPUs.
2. Calculate the **total clock cycles** for both CPUs.